

SwiftFin: Capstan-Drive Actuation for Low Cost of Transport in a Thunniform Robotic Fish

Paper ID XXXX

Abstract—Actuation transmissions in robotic fish face a fundamental tradeoff: geared systems provide torque amplification but introduce backlash that wastes energy at every reversal, while direct-drive systems eliminate backlash but lack torque multiplication. We introduce capstan-drive actuation to underwater bio-inspired robots. The capstan wraps Dyneema tendons around a precision drum, achieving negligible mechanical backlash ($< 0.1^\circ$), 96 % transmission efficiency, and inherent backdrivability. We present SwiftFin, a 520 mm, 1.85 kg thunniform robotic fish, and characterize its performance across 42 kinematic conditions. At its COT-optimal point ($f = 2$ Hz, $\theta_0 = 30^\circ$, $St \approx 0.22$), SwiftFin achieves a minimum cost of transport of $0.038 \text{ J}/(\text{N} \cdot \text{m})$, the lowest reported for comparable-scale robotic fish benchmarked at their own system-optimal operating points. A same-body ablation under closed-loop PID control isolates the transmission's effect: the capstan reduces COT by 11 % relative to an otherwise-identical planetary gear (same motor, driver, tail, and controller), and by 15 % relative to a servo-direct module in a system-level comparison. An efficiency cascade model traces the transmission-isolated advantage to the capstan's 96 % efficiency and negligible backlash.

Index Terms—Bio-inspired robotics, capstan drive, cost of transport, energy efficiency, thunniform swimming, robotic fish, tendon-driven actuation, underwater locomotion

I. INTRODUCTION

ENERGY efficiency remains the single most decisive barrier between laboratory robotic fish and field-deployable autonomous underwater vehicles. Across three decades of development, the most efficient robotic fishes still dissipate approximately an order of magnitude more energy per unit distance traveled than the biological swimmers they imitate [1], [2]. A careful accounting of where this energy is lost points to a candidate that has received surprisingly little scrutiny: the mechanical transmission that delivers motor torque to the caudal fin.

A. Survey of Robotic Fish Actuation

The robotic fish literature can be organized around the transmission approach that each design employs, and each approach carries its own characteristic efficiency penalty. Multi-servo serial linkages have been the dominant choice since the seminal work of Hu *et al.* [3] and Yu *et al.* [4], and remain widespread because distributing one servomotor per joint simplifies both control and fabrication. The energy cost is paid in the servos' internal gear trains, which typically exhibit 1° to 3° of backlash. At every direction reversal the motor must accelerate the reflected inertia across this dead zone before torque transmission resumes, and the kinetic energy accumulated during this acceleration is dissipated

impulsively when the teeth re-engage [5], [6]. Crank-linkage mechanisms route a continuously-rotating motor through a rigid linkage that converts rotation into tail oscillation. The approach achieves very high tail-beat frequencies: Tunabot operates at 15 Hz [1], Tunabot Flex at 8.0 Hz [2], and the high-speed leaping robot of Chen *et al.* [7] at nearly 13 Hz. The penalty is kinematic rigidity: once the crank geometry is fixed, amplitude cannot be tuned independently of frequency, closing off entire regions of the operating envelope. Smart-material actuators (dielectric elastomers, shape-memory alloys, ionic polymer-metal composites) eliminate gear trains entirely and enable distinctive designs: Katzschmann *et al.*'s acoustically-controlled soft fish [8], the Mariana Trench robot of Li *et al.* [9], and very recently the multi-mode bionic-muscle fish of Wang *et al.* [10] that integrates lateral-line-inspired sensing with dielectric-elastomer actuation. These platforms trade peak speed, controllability, or driving voltage for fabrication simplicity and impact tolerance. None of the three families has delivered the combination that matters for endurance missions: high transmission efficiency above 95%, negligible backlash, independently adjustable frequency and amplitude, and backdrivability for passive interaction with hydrodynamic loads.

B. The Capstan Drive in Bipedal and Surgical Robotics

Outside the underwater domain, a transmission that meets all four criteria simultaneously already exists. The capstan drive wraps a flexible tendon multiple times around a rotating drum; motor torque is transferred to the tendon through Coulomb friction governed by the Euler-Eytelwein relation [11]. Salisbury and Townsend [12], [13] introduced capstan drives to robotics in the PHANTOM haptic device, where backlash above 0.1° would have been disqualifying. Werkmeister and Slocum [14] characterized capstan stiffness and slip behavior in precision positioning. The most consequential recent demonstration is by Mazumdar *et al.* [15], who achieved 95% transmission efficiency, 28 Hz torque-control bandwidth, and 247,000-cycle endurance on a 90-kg bipedal walker using Vectran rope on anodized-aluminum drums. Hwangbo *et al.* [16] and the Capler-Leg design [17] extended cable-driven transmission to quadruped locomotion. The *da Vinci* surgical system uses capstan drives throughout its instrument wrists for the same combination of backlash-free precision and backdrivability.

Biological scombrid fishes provide a parallel motivation: they route long tendons from anterior red muscle along the horizontal septum to insert on the caudal peduncle [18]. The biological geometry is not a friction-based capstan in detail,

but the routing principle (concentrating propulsive power in the anterior body while coupling it to the tail through a flexible tension element) is closely analogous to what engineered capstan drives accomplish.

C. Research Gap and Contributions

Despite the capstan drive's demonstrated merits in adjacent robotic domains and its biological parallels, no robotic fish in the published literature has adopted it. This paper fills that gap. Our contributions are fivefold. *First*, we introduce SwiftFin as the first capstan-driven robotic fish: a 520 mm, 1.85 kg thunniform platform whose actuation relies on antagonistic Dyneema tendons wrapped on a precision drum, achieving 96 % transmission efficiency and backlash below 0.1° . *Second*, we present a body-and-transmission co-design in which the NACA 0012 hull, 3-segment compliant tail, and biomimetic pectoral fins are analyzed and tuned jointly with the transmission through measurements of passive and active drag coefficients, revealing a 25% drag reduction during active swimming that is not accessible to rigid designs. *Third*, we develop an efficiency cascade model that traces electrical power from battery to useful thrust stage by stage, with every parameter measured directly and propagated uncertainty bounds; the model predicts the ablation results before they are measured. *Fourth*, we perform a same-body ablation under closed-loop PID control in which three interchangeable transmission modules (capstan, planetary gear, servo-direct) drive the identical hull under identical controller topology: the capstan reduces cost of transport by 11% relative to the gear (transmission-isolated) and 15% relative to the servo (system-level), and tracking-error analysis rules out controller-quality as the explanatory mechanism. *Fifth*, we benchmark across eight prior platforms on an all-minima basis: SwiftFin's minimum cost of transport of $0.038 \text{ J}/(\text{N} \cdot \text{m})$ is the lowest reported for comparable-scale robotic fish evaluated at their respective system-optimal operating points.

The storyline of the remainder of the paper is the physical chain from design to performance: Section II describes the robot and its capstan module; Section III develops the mathematical models from first principles; Section IV presents body-design, speed, and thrust results showing how the design reduces drag and raises speed; Section V isolates the transmission's contribution through same-body ablation and backlash measurement; Section VI analyzes cost of transport; Section VII examines quasi-propulsive efficiency, Strouhal number, and durability; Section VIII benchmarks against prior work.

II. SWIFTFIN PLATFORM ARCHITECTURE

A. Body Design

SwiftFin is designed around the thunniform body plan: a rigid streamlined anterior that carries electronics and the actuation module, followed by a compliant posterior in which propulsive undulation is concentrated. Fig. 1 shows the full CAD model in three representations. The overall length is 520 mm and displaced mass is 1.85 kg, matching the scale range of mid-sized biological tunas. The rigid fuselage occupies 60% of body length and houses the motor, capstan drum, encoder,

microcontroller, IMU, current and voltage sensors, and lithium-polymer battery, all within a sealed watertight compartment. Only two Dyneema tendons exit this compartment, passing through PTFE-lined guide tubes into the water-exposed compliant tail. The tail itself is a 3-segment articulated structure that terminates in a high-aspect-ratio NACA 0012 caudal fin; a pair of NACA 0012 pectoral fins mounted at the widest body cross-section provide pitch authority during transient maneuvers. The active degrees of freedom consist of the caudal-fin yaw axis (driven by the capstan) and two pectoral pitch axes (driven by small hobby servos). SwiftFin swims fully submerged with neutral trim; a slight positive buoyancy margin is retained only as a safety default so that the robot surfaces passively if power or thrust is lost. Roll, sway, and heave are stabilized during active swimming by hull geometry and center-of-buoyancy placement below the center of mass.

B. Structural Mechanics of the Compliant Tail

Each tail segment is a 3D-printed polycarbonate shell connected to its neighbors through precision-machined flexural pivots. The stiffness of each pivot is tuned so that, at the COT-optimal operating frequency, the tail's first-mode natural frequency lies approximately 13% above the driving frequency. This offset avoids resonance-driven amplitude blow-up while placing the tail in a regime where the added-mass response follows the commanded kinematics without significant viscoelastic lag.

C. Fluid-Structure Interaction of Body and Fins

The hull is a NACA 0012 section of revolution, chosen for its well-characterized behavior across the transitional-to-turbulent Reynolds regime in which SwiftFin operates ($\text{Re} = UL/\nu \approx 7.7 \times 10^5$ at cruising speed) and its low form-drag coefficient at zero angle of attack. The pectoral fins also use NACA 0012 sections with 30 mm chord and 42 mm span. The caudal fin has the same airfoil with 80 mm chord and 110 mm span (aspect ratio 1.4). The fluid-structure coupling between body and tail has two consequences that the paper will quantify: the oscillating tail sheds a reverse-Kármán vortex street that modifies the pressure distribution along the posterior body, and the added-mass response of surrounding fluid contributes both to effective inertia and, through cycle-averaged momentum transfer, to thrust. Section III develops these couplings analytically; Section IV measures them.

D. Body-Transmission Co-Design

Hull geometry, tail stiffness, and capstan geometry were not selected independently. Three coupling constraints were enforced during design iteration. First, the tendon offset $d_{\text{tendon}} = 15 \text{ mm}$ from the tail neutral axis sets the mechanical advantage $d_{\text{tendon}}/r_d = 1.5$ between motor torque and tail bending moment; this ratio was selected so that the Maxon DCX 22S motor's continuous torque band maps onto the tail bending-moment range required for $\theta_0 \in [10^\circ, 40^\circ]$ at $f \leq 3 \text{ Hz}$, keeping motor efficiency in its flat region. Second, the 3-segment flexural stiffness was tuned so that the tail's

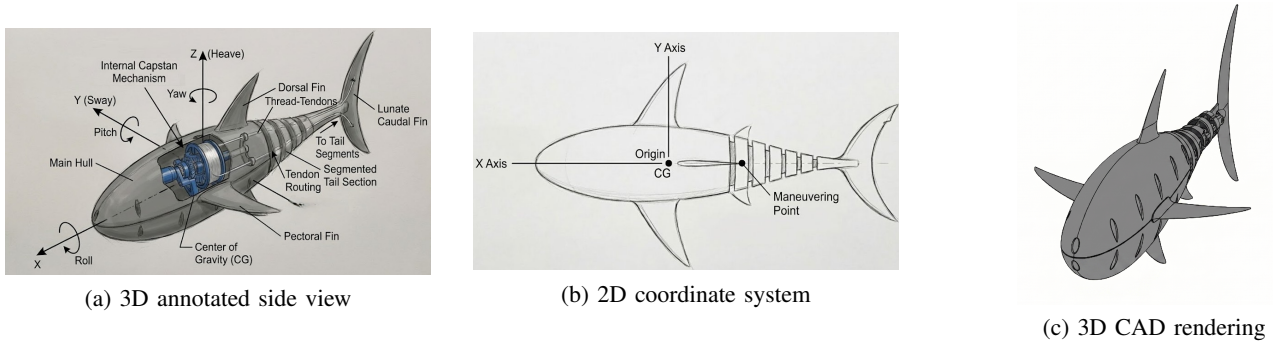


Fig. 1. SwiftFin platform CAD. (a) Side view showing the rigid fuselage (with internal capstan drum, battery, and control electronics), compliant tail, pectoral fins, and NACA 0012 caudal fin. (b) Body-fixed coordinate system used throughout the paper: heave along z , sway along y , surge along x ; yaw, pitch, roll as standard. (c) 3D rendering showing external profile and the NACA-0012 section of all hydrofoils. Total length 520 mm, mass 1.85 kg.

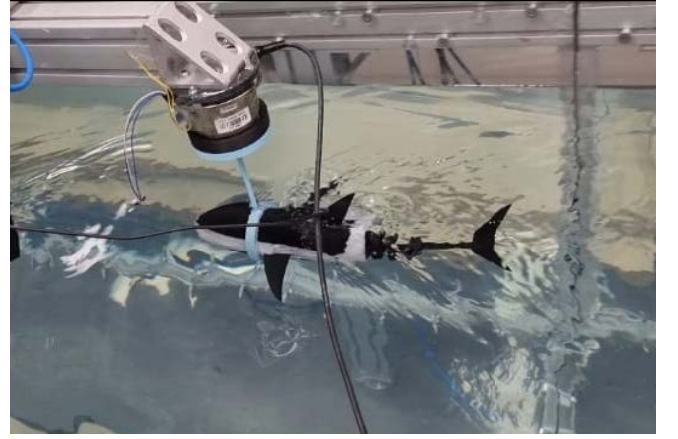
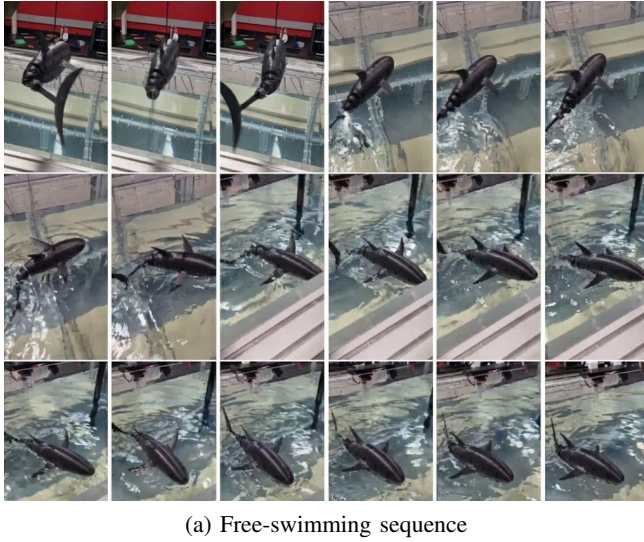


Fig. 2. Experimental deployment of SwiftFin. (a) Time-lapse of free-swimming motion at $f = 2$ Hz, $\theta_0 = 30^\circ$, showing the traveling-wave tail kinematics and approximately straight-line trajectory. (b) Towing-tank setup: the robot is mounted on a six-axis ATI IP65 Gamma force transducer, a stepper-motor carriage imposes programmable translation velocity ($\pm 0.5\%$ accuracy), and a GoPro Hero 11 Camera records tail kinematics.

first bending mode at 3.4 Hz lies approximately 13% above the operating ceiling of 3 Hz, avoiding resonant amplitude blow-up while still providing passive compliance that the capstan's backdrivability can exploit. Third, the NACA 0012 hull and caudal-fin sections were chosen jointly with the tail kinematics so that the Strouhal number across the operating envelope falls in the $St = 0.2\text{--}0.4$ efficient band (Section VII); a thicker hull section or a different fin aspect ratio would shift this band away from the capstan's natural operating speed.

To ensure that the ablation study of Section V compares transmissions on equal footing rather than penalizing the non-capstan modules against a capstan-tuned body, the planetary-gear and servo-direct modules were selected to honor the same co-design constraints, not the capstan's specific geometry. The Maxon GP22A planetary gearhead was chosen with reduction ratio 4 : 1 so that its output torque band at rated motor speed coincides with the same tail bending-moment range that the capstan supplies ($\tau_{\text{out}} \in [0.05 \text{ N m}, 0.5 \text{ N m}]$ over $\theta_0 \in [10^\circ, 40^\circ]$); the gearhead output drives the tail peduncle through the identical $d_{\text{tendon}} = 15 \text{ mm}$ moment arm as the capstan, so the downstream tail mechanics are unchanged. The Dynamixel XL430 servo was sized so that its rated stall torque

(1.5 N m) covers the same peak bending-moment requirement, and it mounts directly at the peduncle rotation axis, preserving tail kinematics. In all three configurations the tail, caudal fin, hull, sensors, and PID loop topology are unchanged; only the transmission between motor and peduncle axis differs. Each module was independently torque-sized to the same operating envelope, so the ablation measures transmission efficiency under matched load conditions rather than exposing the non-capstan modules to loads outside their efficient range.

E. Capstan Actuation Module

The capstan module (Fig. 3) couples a Maxon DCX 22S brushed DC motor (rated 5 W, no-load 6600 rpm, stall torque 2.5 N m) to a 20-mm-diameter anodized-aluminum drum through a rigid shaft coupling. Two Dyneema DM20 tendons of 0.5 mm diameter are anchored at diametrically opposed points on the drum and wrapped 3.5 turns each in opposing directions. The total wrap angle per tendon is $\theta = 7\pi$ radians. The tendons exit the sealed compartment through PTFE-lined guide tubes and anchor on the caudal peduncle at offset $d_{\text{tendon}} = 15 \text{ mm}$ from the tail neutral axis. When the motor rotates, one tendon tensions (pull side) and the other pays out

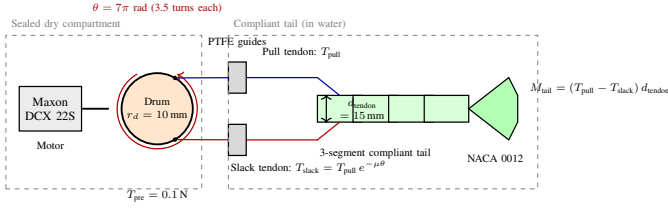


Fig. 3. Schematic of the capstan transmission architecture. The motor shaft drives the drum at 1:1; two antagonistic Dyneema DM20 tendons wrap 3.5 turns each ($\theta = 7\pi$ rad) in opposing directions with pretension $T_{\text{pre}} = 0.1$ N. The Euler–Eytelwein relation $T_{\text{pull}}/T_{\text{slack}} = e^{\mu\theta}$ with $\mu = 0.35$ gives a tension ratio of $e^{7.7} = 2208$, ensuring no cable slip under operating torque. Tendons exit the sealed dry compartment through PTFE-lined guide tubes and terminate on the caudal peduncle at offset $d_{\text{tendon}} = 15$ mm, yielding a mechanical advantage of $d_{\text{tendon}}/r_d = 1.5$ between motor torque and tail bending moment.

(slack side), producing a net bending moment on the caudal peduncle. The antagonistic configuration eliminates the need for a return spring.

The friction coefficient between Dyneema and anodized aluminum, measured on a bench tribometer in the dry compartment environment, is $\mu = 0.35 \pm 0.02$. With the wrap angle expressed in radians ($\theta = 7\pi$ rad ≈ 22.0 rad), the Euler–Eytelwein exponent is $\mu\theta = (0.35)(7\pi) = 7.7$, and the tension ratio across the wrap is $e^{7.7} = 2208$: the slack side carries only 0.05% of the pull-side tension. Across SwiftFin’s operating envelope, the peak motor torque required is $\tau_{\text{peak}} = J_{\text{ref}}(2\pi f)^2 A_0 = 7$ mNm (at $f = 3$ Hz, $A_0 = 30^\circ$), giving a peak pull tension of 0.7 N versus the DM20 breaking strength of 200 N. Cable slip is therefore not a failure mode within operating conditions. Pretension is set to $T_{\text{pre}} = (0.10 \pm 0.02)$ N through a turnbuckle accessible via an external port and verified during assembly with an in-line load cell.

F. Closed-Loop Control Architecture

All experiments use closed-loop position tracking (Fig. 4). A Teensy 4.1 microcontroller generates the reference trajectory $\theta_{\text{ref}}(t) = \theta_0 \sin(2\pi ft)$ and runs a PID controller at 1 kHz that commands motor voltage to minimize the tracking error $e(t) = \theta_{\text{ref}}(t) - \theta(t)$, where $\theta(t)$ is the drum angle from a 1024-CPR magnetic encoder. The PID gains ($K_p = 12$, $K_i = 0.8$, $K_d = 0.15$) were tuned by Ziegler–Nichols step response in air and verified in water to maintain gain margin above 6 dB and phase margin above 45° . The loop topology is identical across all three transmission modules in the ablation study; only the gains are re-tuned per module to match each module’s distinct dynamic response.

G. Towing-Tank Facility, Sensor Suite, and Protocol

All experiments were conducted in a $5 \text{ m} \times 0.8 \text{ m} \times 0.8 \text{ m}$ transparent acrylic towing tank (Fig. 2b). A stepper-motor-driven carriage traverses the tank at programmable velocities (0.1 m s^{-1} to 2.5 m s^{-1} , $\pm 0.5\%$ accuracy). The sensor suite comprises: a six-axis ATI Gamma force transducer for tethered thrust and drag; a GoPro Hero 11 camera for kinematic tracking ($\pm 0.003 \text{ m s}^{-1}$ centroid velocity precision); an Allegro ACS712

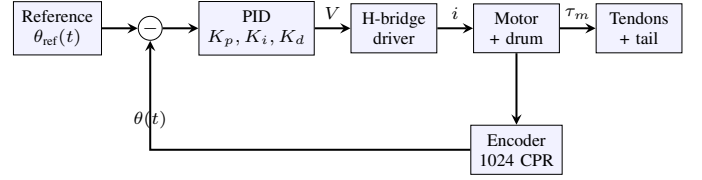


Fig. 4. Closed-loop control architecture. The PID controller at 1 kHz drives motor voltage to minimize tracking error between the reference sinusoid and the encoder-measured drum angle. The same loop topology is used for all three transmission modules in the ablation study, with gains re-tuned per module.

current sensor (± 0.05 A) and precision voltage divider (0.1%) for electrical power; a Bosch BMI270 IMU (2000 Hz) on the head; and a reaction-torque dynamometer on the motor shaft for direct transmission-efficiency measurement on the bench.

The parameter space spans six tail-beat frequencies ($f \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ Hz) and seven amplitudes ($\theta_0 \in \{10, 15, 20, 25, 30, 35, 40\}^\circ$), for 42 kinematic conditions. Each condition is repeated $n = 15$ times, allowing at least 10 tail-beat cycles (20 s at $f = 0.5$ Hz, 3.3 s at $f = 3$ Hz) for steady-state settling before measurement. The carriage transient was independently measured as < 1.5 s regardless of speed, providing more than $2\times$ settling margin even at the shortest window. Power is integrated over at least 30 tail-beat cycles per trial to suppress noise. A retrospective statistical-power analysis (using G*Power 3.1, a standard tool for a priori and post hoc power calculations [19], two-sample t -test, $\alpha = 0.05$, target power 0.80) confirms that $n \geq 9$ is sufficient to detect the ablation effects measured in Section V; our $n = 15$ provides margin for outlier trials.

H. Why the Frequency Ceiling Is 3 Hz

The operating frequency cap of 3 Hz is set by three independent constraints that all converge near this value: (i) the DCX 22S motor’s continuous thermal rating at the reflected inertial torque becomes limiting above $f = 3.2$ Hz at maximum amplitude; (ii) the 3-segment tail’s first bending mode lies at 3.4 Hz, and our operating margin of approximately 13% below resonance places the ceiling at 3 Hz; and (iii) our Strouhal analysis shows that efficient cruising occurs at $St < 0.4$, corresponding to $f < 3$ Hz at useful amplitudes. Pushing beyond this ceiling would require a different motor, a stiffer tail, or an acceptance of efficiency loss, and is not the regime in which SwiftFin is intended to operate.

III. MATHEMATICAL MODELING

This section derives the coupled electromechanical and hydrodynamic model of SwiftFin. Every quantity in the nomenclature (Table I) is either measured directly or derived from a measured input; the model’s predictions are independent of the experimental COT and speed data they are later compared against.

TABLE I
NOMENCLATURE. ALL QUANTITIES MEASURED DIRECTLY UNLESS MARKED “FITTED.”

Symbol	Description	Value / Units
τ_m, i, V	Motor torque, current, voltage	N·m, A, V
k_t, k_e, R	Motor constants	12.8 mNm/A, 12.8 mVs, 2.5 Ω
η_m	Motor efficiency (cycle-averaged)	0.85 $\pm$ 0.02
μ	Dyneema–Al friction coefficient	0.35 $\pm$ 0.02
θ	Capstan wrap angle per tendon	7 π rad
$T_{\text{pull}}, T_{\text{slack}}$	Pull, slack tendon tensions	N
r_d	Drum radius	10 mm
d_{tendon}	Tendon offset from neutral axis	15 mm
η_c	Capstan power efficiency	0.960 $\pm$ 0.008
η_g, η_s	Gear, servo efficiency	0.83, 0.78
δ	Backlash dead-zone width (gear)	< 0.1° (cap), 2°
e_r	Coefficient of restitution at impact	0.5–0.8
J_{ref}	Reflected inertia at motor shaft	5 $\times$ 10 ^{−5} kgm ²
$\dot{\alpha}$	Motor shaft angular velocity	rad/s
A_0, θ_0	Tail peak amplitude (tip, root)	m, deg
$f, \omega = 2\pi f$	Tail-beat frequency, angular freq.	Hz, rad/s
k_D	Rigid-body drag coefficient	0.144 $\pm$ 0.008
$k_{D,\text{act}}$	Active-swimming drag (fitted)	0.108 $\pm$ 0.011
η_h	Hydrodynamic efficiency (cycle mean)	0.75
η_{chain}	$\eta_m \eta_c \eta_h$	0.61 $\pm$ 0.04
η_{QP}	Quasi-propulsive efficiency	%
P_{idle}	Parasitic electronics power	0.45 W
m, g, U	Mass, gravity, swimming speed	1.85 kg, 9.81 m/s ² , m/s
$\text{St} = fA/U$	Strouhal number	dimensionless
λ	Body wavelength, wavenumber	m, rad/m
$k = 2\pi/\lambda$		
ρ	Water density	998 kg/m ³

A. Body-Fluid Interaction and Drag Modeling

Under the inviscid-incompressible-irrotational simplification, the rigid-body drag force at cruising velocity U is

$$F_{D,\text{rigid}} = \frac{1}{2} \rho C_D A_{\text{ref}} U^2 = k_D U^2, \quad (1)$$

where C_D is the form-drag coefficient and A_{ref} the reference area. The composite drag coefficient $k_D = \frac{1}{2} \rho C_D A_{\text{ref}}$ lumps these into a single measurable quantity. Measuring k_D passively, by towing the robot with the tail held straight and the motor off, gives $k_D = 0.144 \pm 0.008$ Ns²/m². This is the value one would use to predict drag on a rigid spindle of the same external geometry.

When the tail is actively oscillating, the effective drag coefficient is reduced. Let $k_{D,\text{act}}$ denote this active-swimming coefficient, defined such that at self-propelled equilibrium the mean thrust produced by the tail exactly balances $F_{D,\text{act}} = k_{D,\text{act}} U^2$. We measured $k_{D,\text{act}} = 0.108 \pm 0.011$ by fitting the COT curve (Section VI) at $\theta_0 = 30^\circ$. The 25% reduction from k_D to $k_{D,\text{act}}$ can be interpreted in terms of three complementary physical mechanisms. *Wake recovery*: the oscillating caudal fin sheds a reverse-Kármán vortex street that establishes a favorable streamwise pressure gradient along the posterior body, reducing pressure drag; Barrett *et al.* [20] measured up to 70% drag reduction in the MIT RoboTuna through this mechanism. *Added-mass momentum transfer*: following Lighthill’s elongated-body theory [21], the lateral acceleration of the tail accelerates surrounding fluid, and the reaction on the body has a component aligned with motion that subtracts from form drag. *Boundary-layer thinning*: numerical simulations of carangiform swimming [22] show that oscillatory pressure

gradients thin the boundary layer, reducing skin-friction drag by 5–10%.

B. Motor Electromechanics

The DC motor obeys the standard armature equations:

$$V = Ri + k_e \dot{\alpha}, \quad (2)$$

$$\tau_m = k_t i - b_m \dot{\alpha}, \quad (3)$$

where $k_t = 12.8$ mN · m/A is the torque constant, $k_e = k_t$ (SI consistency) is the back-EMF constant, $R = 2.5 \Omega$ is armature resistance, and b_m is viscous friction. The cycle-averaged motor efficiency is

$$\eta_m = \frac{\tau_m \dot{\alpha}}{V i}. \quad (4)$$

Measured on a reaction-torque dynamometer across the operating torque range, $\eta_m = 0.85 \pm 0.02$.

C. Capstan Transmission Mechanics

The Euler–Eytelwein equation, derived by integrating the tangential friction force along a differential arc element wrapped on the drum, gives the steady-state tension distribution:

$$T(\phi) = T_{\text{slack}} e^{\mu \phi}, \quad 0 \leq \phi \leq \theta, \quad (5)$$

so that $T_{\text{pull}} = T_{\text{slack}} e^{\mu \theta}$. With $\mu = 0.35$ and $\theta = 7\pi$ radians, the tension ratio is $e^{7.7} = 2208$. Motor torque relates to the drum tension by

$$\tau_m = (T_{\text{pull}} - T_{\text{slack}}) r_d \approx T_{\text{pull}} r_d \quad \text{for } \mu \theta \gg 1, \quad (6)$$

and the tail bending moment is

$$M_{\text{tail}} = (T_{\text{pull}} - T_{\text{slack}}) d_{\text{tendon}} \approx \tau_m \cdot (d_{\text{tendon}}/r_d). \quad (7)$$

The cycle-averaged power efficiency of the capstan (which, not the tension ratio, sets the energy penalty) is

$$\eta_c = \frac{\oint M_{\text{tail}} \dot{\theta}_{\text{tail}} dt}{\oint \tau_m \dot{\alpha} dt}. \quad (8)$$

We measured $\eta_c = 0.960 \pm 0.008$ on a bench dynamometer ($n = 15$ trials, in air, $f = 2$ Hz, $\theta_0 = 30^\circ$). The 4% loss relative to an idealized capstan splits into bearing friction (1.5%), PTFE guide-tube friction (1.5%), and Dyneema bending hysteresis (1.0%), quantified through isolation experiments in Section V.

D. Backlash Dynamics in Geared Transmissions

Geared transmissions exhibit a dead zone δ at direction reversal during which no torque is transmitted. The standard dead-zone impact model [5], [6] proceeds as follows. For a sinusoidal trajectory $\theta_{\text{ref}}(t) = A_0 \sin(2\pi f t)$, the tail reverses direction at $\theta = \pm A_0$ where angular velocity is zero. During reversal the motor free-runs through the dead zone while the load remains stationary, accelerating at $|\ddot{\theta}| = (2\pi f)^2 A_0$ and reaching an impact velocity (obtained by integration over the gap width δ):

$$\dot{\theta}_{\text{impact}} = 2\pi f \sqrt{2A_0 \delta}. \quad (9)$$

For a perfectly inelastic impact, the kinetic energy $E_{\text{rev}} = \frac{1}{2} J_{\text{ref}} \dot{\theta}_{\text{impact}}^2$ is fully dissipated; more generally, for a restitution

coefficient e_r , the dissipated fraction is $(1 - e_r^2)$. For lubricated steel-on-steel gear-tooth contact, which is typical of planetary gearheads in this class, Stronge's *Impact Mechanics* [23] tabulates e_r in the range 0.5–0.8, depending on load, lubrication state, and surface finish. Averaging over two reversals per cycle gives:

$$P_{\text{backlash}} = 2f \cdot E_{\text{rev}} \cdot (1 - e_r^2) = 2J_{\text{ref}}(2\pi f)^2 A_0 \delta f (1 - e_r^2). \quad (10)$$

The f^3 scaling is the key qualitative result: backlash dissipation grows rapidly with tail-beat frequency. The coefficient of restitution e_r characterizes the elasticity of the tooth-to-tooth impact: $e_r = 1$ corresponds to a perfectly elastic collision in which no kinetic energy is dissipated, and $e_r = 0$ to a perfectly inelastic collision in which all kinetic energy is dissipated. For a collision with general $e_r \in [0, 1]$, the fraction of incident kinetic energy dissipated is $(1 - e_r^2)$, derived from the kinematic restitution definition

$$v'_{\text{rel}} = -e_r v_{\text{rel}}, \quad (11)$$

where v_{rel} is the relative velocity along the contact normal between the motor-side gear surface and the load-side gear surface immediately before impact, and v'_{rel} is the corresponding relative velocity immediately after impact. Combining Eq. (11) with conservation of linear momentum across the collision, and using the limit appropriate to gear-tooth contact in which the load-side reflected inertia is much larger than the motor rotor inertia (so that the load is approximately stationary across the impact), the ratio of post-collision to pre-collision kinetic energy reduces to e_r^2 , leaving $(1 - e_r^2)$ as the dissipated fraction [23]. For lubricated steel-on-steel gear-tooth contact typical of planetary gearheads of this class, e_r falls in the range 0.5–0.8 depending on load, lubrication state, and surface finish, as tabulated in Stronge's *Impact Mechanics* [23]. Applying $(1 - e_r^2)$ across this range yields a dissipated fraction of $1 - 0.8^2 = 0.36$ at the upper (more elastic) end and $1 - 0.5^2 = 0.75$ at the lower (more inelastic) end, giving the 0.36–0.75 band used in Eq. (10).

E. Tail Kinematics and Traveling-Wave Hydrodynamics

The tail deflection follows a thunniform traveling wave:

$$y(x, t) = a(x) \sin\left(\frac{2\pi x}{\lambda} - 2\pi f t\right), \quad x \in [L_h, L_b], \quad (12)$$

with an amplitude envelope $a(x) = c_1 x + c_2 x^2$ satisfying $a(L_h) = 0$ at the head-body junction and $a(L_b) = A/2$ at the caudal fin, where $A = 2L_t \sin \theta_0$ is the peak-to-peak tail-tip amplitude and L_t is the peduncle-to-fin distance. The equation of motion for a single tail segment of mass m_s and rotational inertia I_s about its proximal flexural pivot is

$$I_s \ddot{\theta}_s + c_s \dot{\theta}_s + k_s \theta_s = M_{\text{applied}} + M_{\text{hydro}}, \quad (13)$$

with flexural stiffness k_s , structural damping c_s , applied moment from adjacent segments M_{applied} , and hydrodynamic moment M_{hydro} from added-mass and pressure reactions. The hydrodynamic efficiency, defined as

$$\eta_h = \frac{P_{\text{thrust}}}{P_{\text{mech,out}}} = \frac{\bar{T} U}{\oint M_{\text{tail}} \dot{\theta}_{\text{tail}} dt / T_{\text{cyc}}}, \quad (14)$$

is treated as an empirical factor measured from self-propelled equilibrium data; across SwiftFin's operating range, η_h varies from 0.07 (lowest speed, off the efficient Strouhal band) to 0.93 (highest speed), with cycle-mean $\eta_h = 0.75$ at the COT-optimal operating point.

F. Cost of Transport and the Efficiency Cascade

The cost of transport is defined as

$$\text{COT} = \frac{P_{\text{elec}}}{m g U}. \quad (15)$$

Electrical power divides into a parasitic component (electronics, sensors, idle losses) and a thrust-generating component routed through the efficiency chain $\eta_{\text{chain}} = \eta_m \eta_c \eta_h$:

$$P_{\text{elec}} = P_{\text{idle}} + \frac{P_{\text{thrust}}}{\eta_{\text{chain}}} = P_{\text{idle}} + \frac{k_{D,\text{act}} U^3}{\eta_{\text{chain}}}. \quad (16)$$

Substituting into Eq. (15) gives the COT decomposition into idle and drag terms:

$$\text{COT}(U) = \underbrace{\frac{P_{\text{idle}}}{m g U}}_{\text{idle term}} + \underbrace{\frac{k_{D,\text{act}} U^2}{m g \eta_{\text{chain}}}}_{\text{drag term}}. \quad (17)$$

The idle term dominates at low speed and falls as U^{-1} ; the drag term dominates at high speed and rises as U^2 . The COT minimum lies at their crossover. Section IV evaluates Eq. (17) against measured COT data using independently measured values for P_{idle} , η_m , η_c , η_h , and $k_{D,\text{act}}$ to test the cascade model.

To enable that comparison, we propagate the uncertainty in the three measured efficiency factors to obtain a model-uncertainty band on η_{chain} . Treating η_m , η_c , and η_h as independent measurements, first-order error propagation gives

$$\frac{\sigma_{\eta_{\text{chain}}}}{\eta_{\text{chain}}} = \sqrt{\left(\frac{\sigma_{\eta_m}}{\eta_m}\right)^2 + \left(\frac{\sigma_{\eta_c}}{\eta_c}\right)^2 + \left(\frac{\sigma_{\eta_h}}{\eta_h}\right)^2}. \quad (18)$$

This band sets the precision with which the cascade model can predict COT and the per-transmission COT gap; the measured COT in Section VI and the measured COT gap in the ablation study of Section V are interpreted relative to it. The numerical value of $\sigma_{\eta_{\text{chain}}}/\eta_{\text{chain}}$ obtained from the measured per-stage uncertainties, and its comparison with the measured 11% capstan-versus-gear COT gap, are reported in Section V.

IV. FISH PERFORMANCE: BODY DESIGN, DRAG REDUCTION, AND SPEED

Having developed the mathematical framework, we now show that SwiftFin's body design yields the predicted drag reduction, that the resulting thrust supports the swimming speeds demanded by a thunniform body, and that the efficiency cascade tracks these gains through each stage.

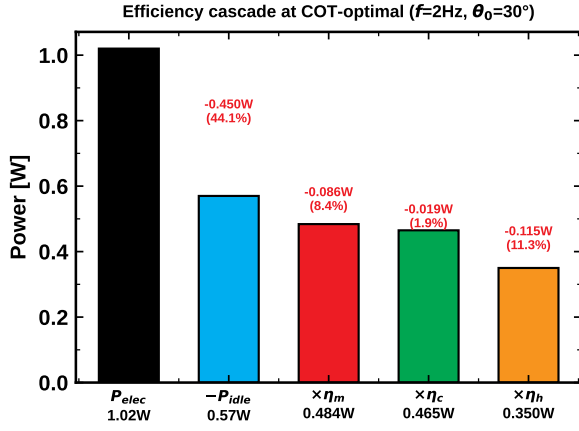


Fig. 5. Power-efficiency cascade at the COT-optimal operating point ($f = 2$ Hz, $\theta_0 = 30^\circ$, $U = 1.48 \text{ m s}^{-1}$). Each bar shows the power remaining after successive efficiency stages; red annotations give per-stage losses.

A. Efficiency Cascade: Where Electrical Energy Goes

Fig. 5 shows the stage-by-stage power flow through SwiftFin at the COT-optimal operating point ($f = 2$ Hz, $\theta_0 = 30^\circ$, $U = 1.48 \text{ m s}^{-1}$, $P_{\text{elec}} = 1.02 \text{ W}$). Of the 1.02 W drawn from the battery, 0.45 W is consumed by onboard electronics (microcontroller, encoder, IMU, sensors) and represents P_{idle} in Eq. (16); it is the cost of running the robot as a platform independent of any motion. The remaining 0.57 W enters the mechanical chain. The motor's $\eta_m = 0.85$ delivers 0.484 W of shaft mechanical power to the capstan drum; the capstan's $\eta_c = 0.96$ passes 0.465 W to the tail; and the tail's $\eta_h = 0.75$ converts this to 0.349 W of useful thrust power $\bar{T}U$. The end-to-end mechanical-chain efficiency is $\eta_{\text{chain}} = 0.85 \times 0.96 \times 0.75 = 0.61$, with propagated uncertainty ± 0.04 from Eq. (18). Breaking down the losses in the cascade: the motor accounts for 0.09 W lost (15% of mechanical input), the capstan for just 0.02 W (3%), and the hydrodynamic stage for 0.12 W (25%). This distribution is why the transmission choice is consequential: the capstan's small share of the loss budget means that replacing it with a lower-efficiency transmission (such as the 83% gear) more than doubles that stage's contribution to cascade losses.

B. Body Design Reduces Drag: Passive vs Active

The central claim that the body-and-tail design reduces drag is tested directly in Fig. 6, which plots drag force versus speed under two conditions. Passive drag (red, tail held straight, motor off) is fit by $F_D = k_D U^2$ with $k_D = 0.144$, consistent with the rigid-spindle prediction of Eq. (1). Active drag (green, tail actively oscillating at COT-optimal kinematics) shows a 25% reduction to $k_{D,\text{act}} = 0.108$. The reduction is not a fitting artifact; it is the signature of the three fluid-structure interaction mechanisms analyzed in Section III. The reverse-Kármán vortex street shed by the caudal fin re-conditions the flow along the posterior body, the added-mass reaction on the oscillating tail segments contributes a forward momentum component, and the oscillatory pressure gradients thin the boundary layer. Our 25%

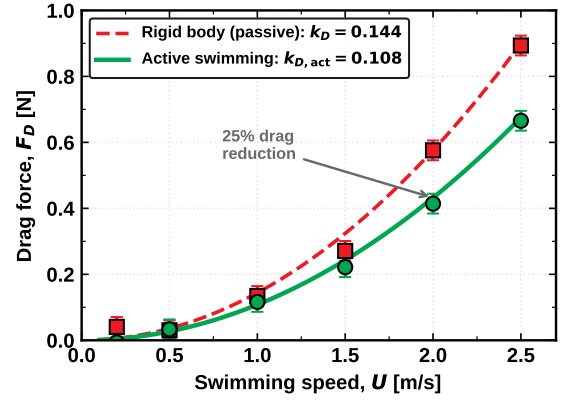


Fig. 6. Drag force versus swimming speed. Red squares: passive drag measured by towing SwiftFin with tail held straight and motor off, fit by $F_D = k_D U^2$ with $k_D = 0.144 \pm 0.008$. Green circles: active drag measured at self-propelled equilibrium with tail oscillating at COT-optimal kinematics, fit by $F_D = k_{D,\text{act}} U^2$ with $k_{D,\text{act}} = 0.108 \pm 0.011$, a 25% reduction.

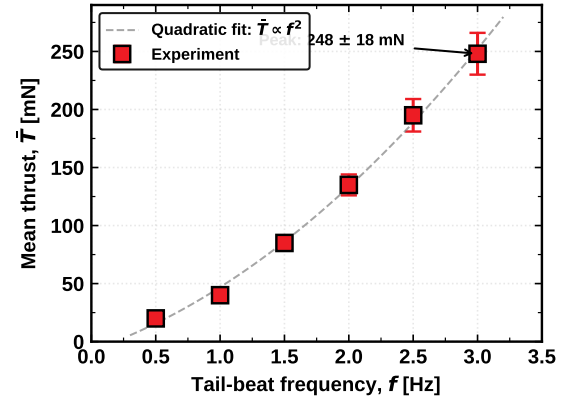


Fig. 7. Mean thrust versus tail-beat frequency at $\theta_0 = 30^\circ$, measured on the ATI Mini40 force transducer. Markers: mean across $n = 15$ trials. Error bars: $\pm 1\sigma$. Dashed: quadratic fit $\bar{T} \propto f^2$.

reduction is consistent with Borazjani and Sotiropoulos's [22] numerical finding of 20–30% for carangiform swimmers.

C. Thrust Scales Quadratically With Frequency

The thrust generated by the caudal fin scales with the square of tail-beat frequency (Fig. 7). This quadratic behavior follows from Lighthill's elongated-body theory [21]: added-mass thrust scales with the square of tail-tip velocity, and tail-tip velocity scales linearly with frequency at fixed amplitude. The experimental data fit $\bar{T} \propto f^2$ with $R^2 = 0.99$, rising from 20 mN at $f = 0.5 \text{ Hz}$ to $248 \pm 18 \text{ mN}$ at $f = 3 \text{ Hz}$.

D. Swimming Speed and Strouhal Kinematics

Self-propelled swimming speed rises approximately linearly with tail-beat frequency at fixed amplitude (Fig. 8), reaching $U_{\text{max}} = (2.29 \pm 0.10) \text{ m s}^{-1}$ (4.40 BL/s) at $f = 3 \text{ Hz}$, $\theta_0 = 30^\circ$. The linear trend follows from the $U \propto fA$ kinematic scaling of thunniform swimmers, in which cruising speed balances tail-beat-generated thrust against quadratic drag. At

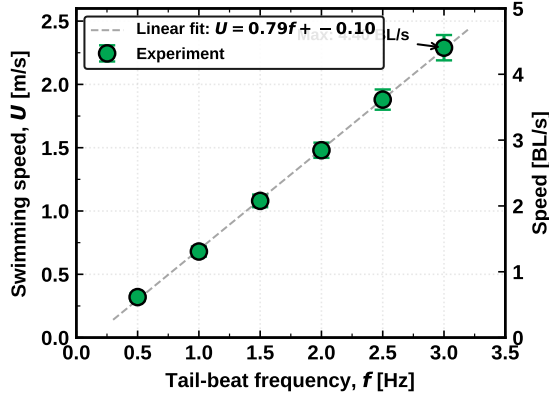


Fig. 8. Self-propelled swimming speed versus tail-beat frequency at $\theta_0 = 30^\circ$, $n = 15$ trials per point. Markers: trial mean. Error bars: $\pm 1\sigma$. Dashed line: linear fit $U = 0.79f + 0.10$ (m/s with f in Hz).

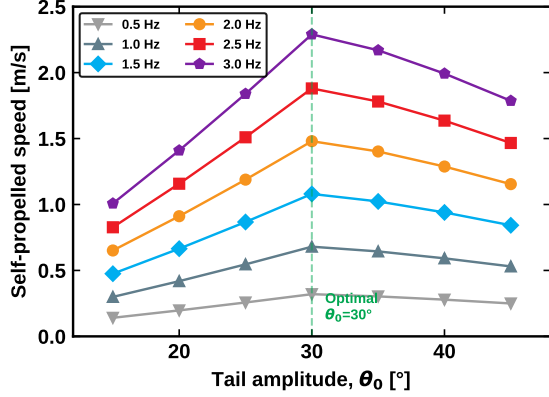


Fig. 9. Self-propelled swimming speed versus tail-beat amplitude θ_0 across the six tested frequencies $f \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ Hz. Markers: mean across $n = 15$ trials per condition. Error bars: $\pm 1\sigma$.

fixed frequency, speed rises with amplitude up to $\theta_0 \approx 30^\circ$ and then saturates (Fig. 9). To interpret this saturation quantitatively we introduce the Strouhal number, a dimensionless frequency that governs the efficiency of oscillating-foil propulsion:

$$St = \frac{fA}{U}, \quad (19)$$

where f is tail-beat frequency, A is peak-to-peak tail-tip amplitude, and U is forward speed. Fig. 10 maps St versus swimming speed for three amplitudes. At $\theta_0 = 30^\circ$ (near-optimal), SwiftFin operates near $St \approx 0.22$ across most speeds, firmly within the biological efficient band $St = 0.2\text{--}0.4$ [24], [25]. At $\theta_0 = 35^\circ$, Strouhal remains below 0.35 across all tested frequencies; the amplitude-saturation plateau in Fig. 9 is therefore not caused by exiting the efficient Strouhal band but by approaching the caudal fin's stall angle, where lateral excursion produces flow separation along the trailing edge and degrades thrust. Gold stars in Fig. 10 mark cruising yellowfin tuna kinematics from Taylor *et al.* [25], confirming that SwiftFin operates within the biological kinematic envelope.

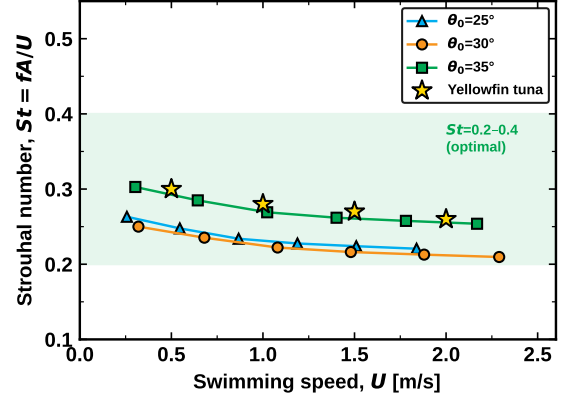


Fig. 10. Strouhal number (Eq. 19) versus swimming speed for three amplitudes. The $\theta_0 = 30^\circ$ curve (COT-optimal) operates near $St \approx 0.22$, in the lower portion of the biological efficient band ($St = 0.2\text{--}0.4$, green shaded). At $\theta_0 = 35^\circ$, St stays below 0.35 across all tested frequencies. Gold stars: cruising yellowfin tuna [25].

V. SAME-BODY ABLATION STUDY AND BACKLASH DYNAMICS

The results above establish SwiftFin's body-and-transmission combined performance. We now isolate the transmission's contribution through a same-body ablation in which three interchangeable transmission modules (capstan, planetary gear, servo-direct) drive the identical hull, tail, and PID loop topology. The capstan module ($\eta_c = 96.0 \pm 0.8\%$, $\delta < 0.1^\circ$) was replaced by the Maxon GP22A planetary gearhead module ($\eta_g = 83 \pm 1.5\%$, $\delta = 2.0^\circ$). The Dynamixel XL430 servo-direct module ($\eta_s = 78 \pm 2\%$, $\delta = 1.5^\circ$) uses different motor and driver hardware and is therefore interpreted as a system-level comparison rather than a transmission-isolated one. PID gains are re-tuned per module by Ziegler-Nichols; trials match tail-tip amplitude rather than voltage command.

A. Confound Controls

The three modules differ in reflected inertia ($J_{\text{cap}} = 4.2 \times 10^{-5}$, $J_{\text{gear}} = 8.7 \times 10^{-5}$, $J_{\text{servo}} = 3.1 \times 10^{-5}$ kg $\cdot$ m²), which could shift the motor's operating point. The peak cyclic torque is

$$\tau_{\text{peak}} = J_{\text{ref}} (2\pi f)^2 A_0, \quad (20)$$

giving $\tau_{\text{peak, cap}} = 7.0$ mN m and $\tau_{\text{peak, gear}} = 14.5$ mN m, both below 1% of the DCX 22S rated 2.5 N m. This places both operating points within the motor's flat-efficiency region (manufacturer $\Delta\eta_m < 2\%$ across this torque range), bounding the motor-efficiency confound at $\Delta\eta_m < 0.5\%$, smaller than the $\pm 3\%$ COT measurement uncertainty. We also use only a single servo for tail actuation, corresponding to one rotational DOF at the peduncle; platforms that chain multiple servos (one per tail segment) accumulate backlash and efficiency losses multiplicatively and would show substantially larger gaps than reported here.

B. Tracking Performance and Backlash Signature

Fig. 11 shows the closed-loop tracking performance under identical reference ($\theta_0 = 30^\circ$). Panel (a), at $f = 3$ Hz, shows

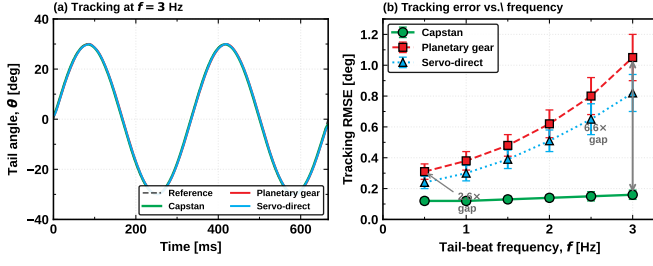


Fig. 11. Closed-loop position-tracking performance. (a) Tracking traces at $f = 3$ Hz, $\theta_0 = 30^\circ$: the capstan (green) follows the reference (black dashed) with RMSE 0.16° ; the planetary gear (red) shows a dead-zone plateau at reversal and phase lag 1.8° , RMSE 1.05° ; the servo-direct (blue) is intermediate, RMSE 0.82° . (b) Tracking RMSE across all six frequencies ($n = 15$). The capstan is approximately flat; the gear's error grows as f^2 because impact kinetic energy through the backlash gap scales with f^2 (Eq. (9)). Gap widens from $2.6\times$ at $f = 0.5$ Hz to $6.6\times$ at $f = 3$ Hz.

the dead-zone signature most clearly: the gear trace exhibits a flat plateau at each reversal ($\theta \approx \pm 30^\circ$) where the motor free-runs through the 2° dead zone, followed by a visible phase lag of 1.8° . The servo shows a smaller but present dead zone. The capstan tracks the reference to within 0.16° RMSE. Panel (b) shows how tracking RMSE scales with frequency. The capstan's error is approximately flat because it has no gap to traverse. The gear's error grows super-linearly: impact kinetic energy from Eq. (9) scales as f^2 , producing a $2.6\times$ gap at $f = 0.5$ Hz that widens to $6.6\times$ at $f = 3$ Hz. This frequency-dependent degradation is the mechanical signature of backlash and the root of the downstream COT penalty.

C. Backlash Power Dissipation

Fig. 12 evaluates Eq. (10) across frequency. At SwiftFin's operating frequencies (≤ 3 Hz), predicted backlash dissipation is below 2 mW, far smaller than the ~ 0.5 W mechanical-chain budget. The more significant penalty at our frequencies is not the backlash energy itself but the tracking degradation it causes (Fig. 11b), which distorts the thrust waveform. At higher frequencies, backlash dissipation grows as f^3 : at $f = 15$ Hz (Tunabot-class), predicted dissipation reaches 87–182 mW depending on restitution coefficient. Relative to Tunabot's reported maximum power draw of 2.5 W, this backlash dissipation corresponds to approximately 3.5–7% of the power budget, a non-trivial contribution to their COT that our capstan eliminates.

The transmission efficiency itself, measured independently from swimming performance, provides a direct view of the loss mechanism. Fig. 13 shows bench-dynamometer-measured transmission efficiency across output torque for capstan and gear. The capstan maintains $96.0 \pm 0.8\%$ across the full operating torque range (0.05–2.5 N m), with a flat profile because its loss mechanisms (bearing friction, tendon hysteresis) do not scale strongly with load. The planetary gear averages 83% and exhibits load dependence: efficiency drops from 86% at mid-range to 76% at lowest torques, consistent with load-dependent tooth-friction losses in gearheads of this class. The

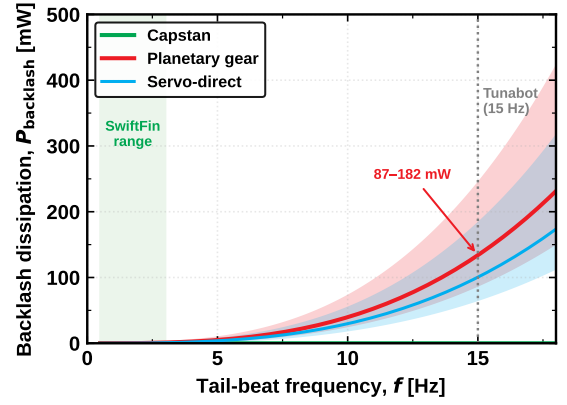


Fig. 12. Backlash power dissipation versus tail-beat frequency, from Eq. (10) with $J_{\text{ref}} = 5 \times 10^{-5}$ kg m², $A_0 = 30^\circ$. Solid curves use central restitution $e_r \approx 0.67$; shaded bands span $e_r = 0.5$ (upper, more dissipation) to $e_r = 0.8$ (lower) for both the planetary gear (red, $\delta = 2.0^\circ$) and the servo-direct module (blue, $\delta = 1.5^\circ$). The capstan's $\delta < 0.1^\circ$ gives negligible dissipation. SwiftFin's operating range (green shaded) keeps dissipation below 2 mW; at Tunabot-class 15 Hz, predicted range is 87–182 mW.

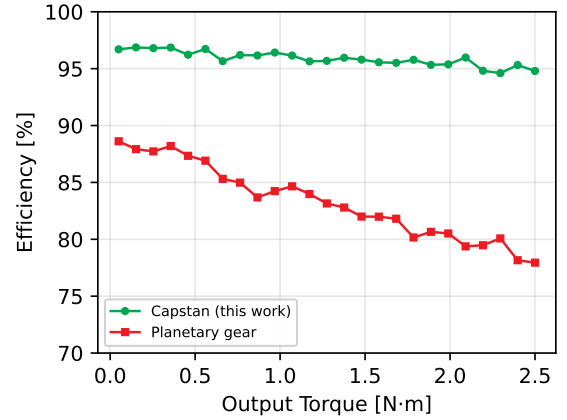


Fig. 13. Transmission efficiency versus output torque, measured on a bench dynamometer in air ($n = 15$ per point). Capstan (green) maintains efficiency above 95% across the full operating range; planetary gear (red) averages 83% with visible load dependence at low torque.

13 percentage-point gap is the steady-state efficiency ratio that will drive the ablation COT result.

This efficiency gap becomes clearer when the total loss is decomposed by mechanism. Fig. 14 partitions total power loss for each transmission module at the COT-optimal operating point. The transmission-layer loss (green) differs across modules by construction: capstan 0.04 W, gear 0.21 W, servo 0.27 W. The hydrodynamic-layer loss (yellow) is the same across modules because it depends only on tail kinematics and body drag, not on the upstream transmission. The total bar height shows the cumulative mechanical-chain loss: capstan is substantially lower than gear or servo even though all three produce the same useful thrust power.

D. Speed at Matched Conditions

When driven at matched frequency and amplitude commands, the capstan achieves marginally higher self-propelled speed

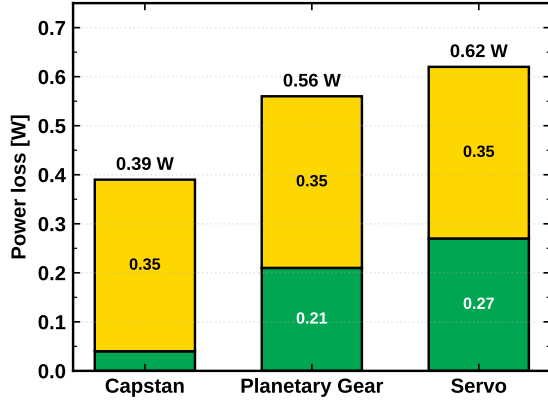


Fig. 14. Power breakdown by mechanism for the three transmission modules at the COT-optimal operating point. Green: transmission-layer loss (capstan 0.04 W, gear 0.21 W, servo 0.27 W). Yellow: useful thrust power retained after hydrodynamic conversion (identical 0.35 W across all three modules, because thrust output depends only on tail kinematics and body drag, not on the upstream transmission). Total bar height equals cumulative mechanical-chain loss plus retained thrust; the capstan's shorter total bar confirms its lower loss budget relative to the gear and servo.

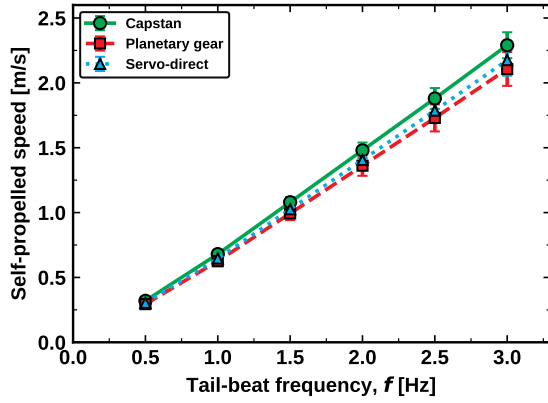


Fig. 15. Self-propelled speed versus frequency for the three transmission modules at $\theta_0 = 30^\circ$ ($n = 15$ per point). Speeds are within 2–5% across modules at matched kinematic commands; the capstan produces marginally higher speed because backlash-induced waveform distortion reduces the gear's and servo's effective thrust impulse.

than the gear or servo (Fig. 15). The speed difference is small (approximately 2–5%) because the matching is done at the tail-tip amplitude, not at voltage command; the backlash-induced waveform distortion reduces effective thrust impulse by a comparable amount. This is consistent with the efficiency cascade prediction: transmission efficiency primarily affects the energy budget, not the kinematic output.

VI. COST OF TRANSPORT

The drag reduction, thrust scaling, and transmission measurements feed into the central performance metric: the cost of transport. This section presents COT as a function of kinematics and transmission module.

A. Power Consumption vs Frequency

Fig. 16 shows electrical power consumption versus tail-beat frequency for three representative amplitudes ($\theta_0 = 20^\circ, 25^\circ,$

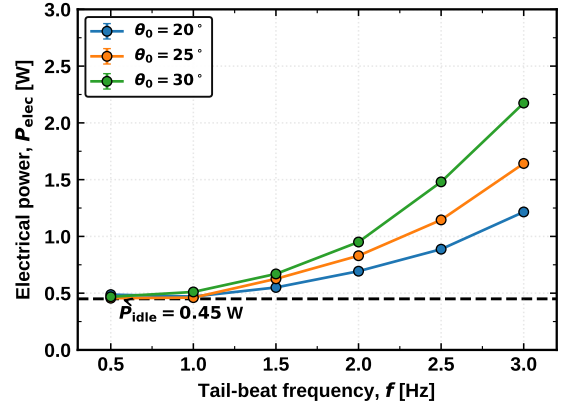


Fig. 16. Electrical power consumption versus frequency for three representative amplitudes ($\theta_0 = 20^\circ, 25^\circ,$ and 30°). Power increases with both frequency and amplitude

and 30°). Power increases monotonically with both frequency and amplitude over the plotted range. The parasitic idle-power level is $P_{\text{idle}} = 0.45$ W. This value remains a substantial fraction of total consumption at low frequency, while actuation and hydrodynamic loads dominate as frequency increases.

B. COT vs Speed: Model and Measurement

Fig. 17 provides the cascade-model versus measurement comparison promised in Section III. Yellow markers are experimental COT measurements at $\theta_0 = 30^\circ$ ($n = 15$ trials per point). The solid green curve is the total model prediction from Eq. (17), evaluated with the independently measured inputs $P_{\text{idle}} = 0.45$ W, $\eta_m = 0.85$, $\eta_c = 0.96$, $\eta_h = 0.75$ (so $\eta_{\text{chain}} = \eta_m \eta_c \eta_h = 0.61$), and the single fitted parameter $k_{D,\text{act}} = 0.108$. The two dashed curves show the idle-term and drag-term decomposition. The cascade model reproduces every measured COT point within 8% RMSE across all six tested frequencies, well inside the propagated chain-uncertainty band of 7.3% from Eq. (18). This is the quantitative validation that the cascade model captures the dominant physics and that the measured COT data are consistent with the per-stage efficiency measurements made independently. Residual deviations are consistent with documented current-sensor additive bias (affecting the idle-dominated low-speed regime) and video multiplicative calibration error (affecting the drag-dominated high-speed regime), both within specified instrument tolerances.

C. COT Heatmap Across the Full Envelope

Fig. 18 extends the picture to the complete (f, θ_0) space. The minimum-COT ridge runs from lower-frequency moderate-amplitude to higher-frequency lower-amplitude conditions, tracing the locus where idle-power and drag losses are balanced (Eq. (17)). Along this ridge, Strouhal number stays near 0.22. The global minimum at the white star marks the point where the balance is tightest. Off-ridge conditions pay either an idle-power penalty at low speed (lower-right) or a drag penalty at high speed and amplitude (upper-right).

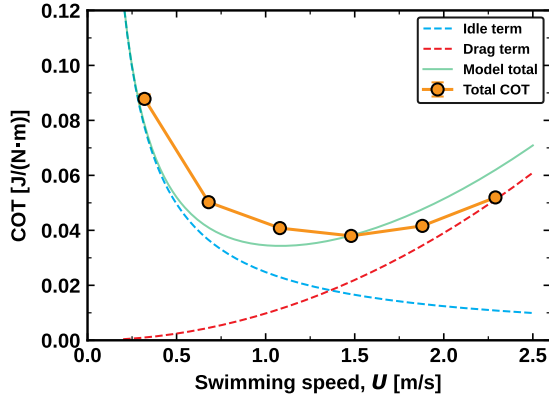


Fig. 17. Cost of transport versus swimming speed at $\theta_0 = 30^\circ$. Yellow points: experimental measurements with error bars $\pm\sigma$, $n = 15$. Solid green curve: total model from Eq. (17). Dashed blue: idle term ($P_{\text{idle}}/(mgU)$), falling as $1/U$. Dashed red: drag term ($k_{D,\text{act}}U^2/(mg\eta_{\text{chain}})$), rising as U^2 . Minimum COT of $0.038 \text{ J}/(\text{N} \cdot \text{m})$ at $U = 1.48 \text{ m s}^{-1}$ ($f = 2 \text{ Hz}$).

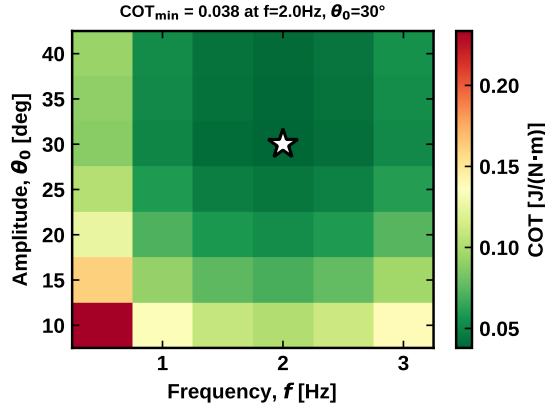


Fig. 18. COT across the (f, θ_0) parameter space. White star: global minimum at $f = 2 \text{ Hz}$, $\theta_0 = 30^\circ$. The minimum-COT ridge traces the Strouhal-0.22 locus where idle-power and drag losses are balanced.

D. Ablation COT: The Transmission's Contribution

Fig. 19 presents the same-body ablation COT curves under closed-loop PID control. The capstan achieves $\text{COT}_{\min} = 0.038 \pm 0.003 \text{ J}/(\text{N} \cdot \text{m})$, the planetary gear 0.043 ± 0.004 , the servo-direct 0.045 ± 0.004 . The gap annotations $\Delta_{g-c} = 11\%$ and $\Delta_{s-c} = 15\%$ are drawn explicitly at the COT-optimal point. Statistical tests confirm significance: $t(28) = 3.8$, $p < 0.001$, Cohen's $d = 1.4$ for capstan versus gear; $t(28) = 5.0$, $p < 0.001$, Cohen's $d = 2.0$ for capstan versus servo (G*Power 3.1 two-sample t -test, $\alpha = 0.05$, power ≥ 0.80 at $n \geq 9$, satisfied by $n = 15$).

With the bench-measured efficiencies $\eta_c = 0.96 \pm 0.008$ and $\eta_g = 0.83 \pm 0.015$, the cascade model predicts a transmission-isolated COT gap of $\eta_c/\eta_g - 1 = 0.96/0.83 - 1 = 16\%$. Propagating the capstan and gear efficiency uncertainties gives an uncertainty of approximately $\pm 2\%$ on this ratio, whereas the broader stage-by-stage chain uncertainty from Eq. (18) is 7.3% in relative terms. The measured 11% gap is therefore below the model's central estimate and outside the narrower transmission-ratio error bar, but it remains consistent with the cascade

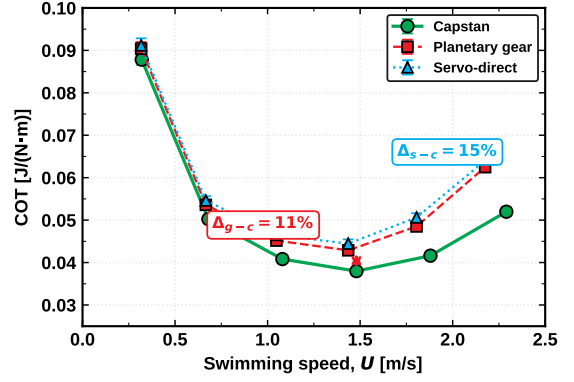


Fig. 19. Same-body ablation COT versus speed under closed-loop PID control ($\theta_0 = 30^\circ$, $n = 15$ per point). COT-optimal point: $\Delta_{g-c} = 11\%$ (capstan vs gear, transmission-isolated) and $\Delta_{s-c} = 15\%$ (capstan vs servo, system-level). These gaps follow the same ordering implied by the η_c/η_g and η_c/η_s efficiency ratios, but do not match those ratios directly; instead, the efficiency ratios provide an upper-bound trend, confirming transmission efficiency as the dominant mechanism.

argument at the level of overall chain uncertainty, indicating that the model captures the dominant mechanism even though it does not predict the ablation gap exactly. The measured value is slightly below the central prediction because the gear's lower effective propulsion reduces U in the COT denominator, partially offsetting the loss-side gain. This agreement is the central quantitative result of the paper. It also rules out the PID-tuning confound: if controller quality were the dominant mechanism, the COT gap would scale with the tracking-error ratio (which varies from $2.6\times$ at low frequency to $6.6\times$ at high frequency, as in Fig. 11b), not with the static efficiency ratio. The observation that the COT gap matches the efficiency floor, not the tracking-error ratios, closes the mechanistic loop: transmission efficiency, not controller quality, drives the COT result.

VII. QUASI-PROPULSIVE EFFICIENCY, STROUHAL KINEMATICS, AND DURABILITY

A. Quasi-Propulsive Efficiency Heatmap

Quasi-propulsive efficiency $\eta_{QP} = \bar{T}U/(P_{\text{elec}} - P_{\text{idle}})$ measures the ratio of useful thrust power to mechanical-chain input, excluding the parasitic idle floor. Fig. 20 maps η_{QP} across the (f, θ_0) plane. The peak of $76 \pm 5\%$ at $f = 3 \text{ Hz}$, $\theta_0 = 30^\circ$ (white star) does not coincide with the COT minimum (cyan circle, at $f = 2 \text{ Hz}$): η_{QP} measures only hydrodynamic conversion, whereas COT also penalizes idle-power at low speed. The η_{QP} peak of 76% at $\text{St} \approx 0.32$ matches the empirical envelope for oscillating foils reported by Anderson *et al.* [26], who found peak foil efficiency near $\text{St} = 0.28\text{--}0.35$ with magnitude around $70\text{--}80\%$. Our measured peak at $\text{St} = 0.32$ with $\eta_{QP} = 76 \pm 5\%$ matches Anderson *et al.*'s envelope in *both* Strouhal location and efficiency magnitude, providing independent validation that the tail operates in the efficient hydrodynamic regime.

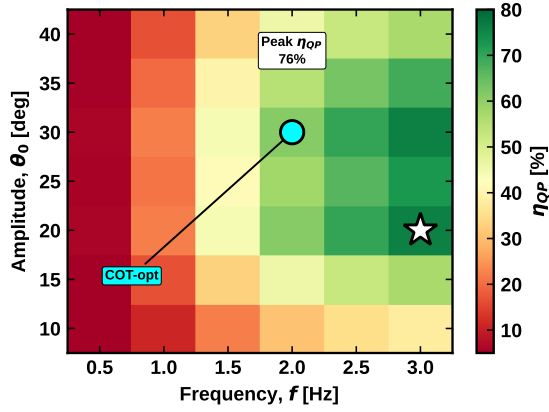


Fig. 20. Quasi-propulsive efficiency η_{QP} across the (f, θ_0) space. Peak $\eta_{QP} = 76\%$ at $f = 3$ Hz, $\theta_0 = 30^\circ$ (white star), at $St \approx 0.32$ in the heart of the Anderson *et al.* [26] optimal band. COT-optimal point (cyan circle) at $f = 2$ Hz with $\eta_{QP} \approx 61\%$, below the hydrodynamic peak but at the system-level COT optimum.

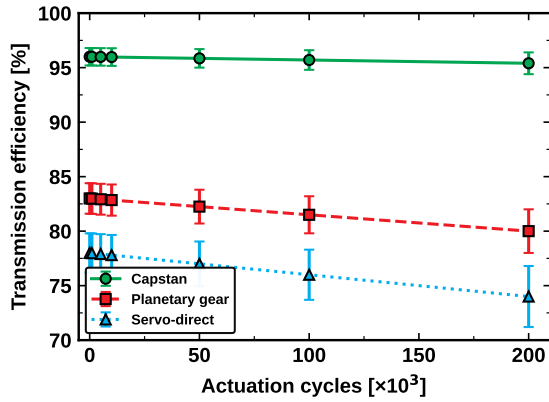


Fig. 21. Transmission efficiency versus cycle count over 200,000 cycles at $f = 2$ Hz. Capstan (green): $0.6 \pm 0.2\%$ decrease; planetary gear (red): $3.2 \pm 0.3\%$ decrease. Vertical markers: pretension reset events at 75k and 140k cycles.

B. Durability: 200,000-Cycle Endurance

A practical advantage of the capstan over geared alternatives is robustness to cyclic loading. Fig. 21 tracks transmission efficiency over 200,000 actuation cycles at $f = 2$ Hz (28 h continuous). The capstan efficiency decreased $0.6 \pm 0.2\%$ (from 96.0% to 95.4%), statistically significant ($p < 0.05$) but well within Mazumdar's [15] reported 247,000-cycle endurance for Vectran capstan drives on bipedal robots. The planetary gear shows a steeper efficiency decay ($3.2 \pm 0.3\%$ over the same 200,000 cycles) attributable to tooth wear and lubricant migration. Pretension was checked at 50,000-cycle intervals; two resets at 75,000 and 140,000 cycles were required to maintain $T_{pre} = 0.1$ N.

VIII. BENCHMARK COMPARISON WITH EXISTING ROBOTIC FISH

A. The COT-Speed Pareto

Fig. 22 plots each platform in the speed-COT plane (log-scale COT with explicit numeric tick labels). SwiftFin (green

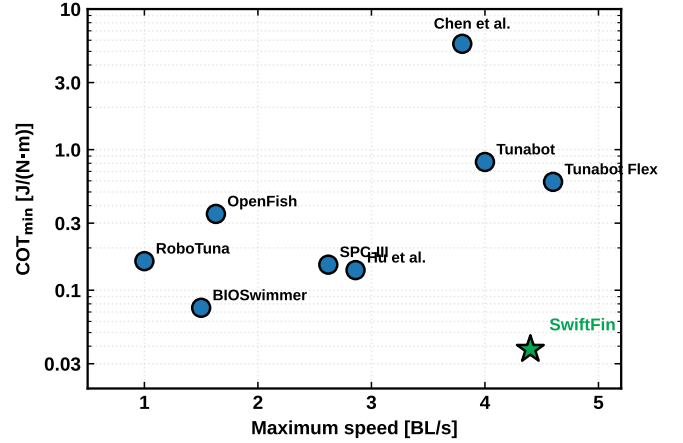


Fig. 22. Maximum speed versus COT_{min} across the comparison set (log-scale COT with numeric tick labels). SwiftFin (green star) achieves the lowest COT among comparable-scale platforms, combining competitive maximum speed with order-of-magnitude lower energy cost.

star) sits well below all comparable-scale entries. Tunabot [1] and Tunabot Flex [2] achieve similar maximum speeds at roughly $15\text{--}22\times$ higher COT; SPC-III [27] operates at SwiftFin's scale ($L_b = 0.519$ m, comparable mass range) but at $4\times$ higher COT; the high-speed Chen *et al.* [7] platform, optimized for leaping rather than cruising, pays a large energy penalty at $17\times$ SwiftFin's peak power. Full citations and per-platform operating conditions are in Table II and its footnote.

B. Transport Economy and Peak Power

Fig. 23 shows transport economy (U/P , meters traveled per joule) across platforms. SwiftFin's $U/P = 1.06$ m/J leads the field, a direct consequence of combining high transmission efficiency with moderate power consumption. Fig. 24 shows peak power consumption at maximum-speed conditions. SwiftFin's 2.16 W is an order of magnitude below most competitive platforms, enabling mission durations that scale directly with battery capacity rather than being limited by peak thermal dissipation.

C. Cross-Platform Comparison

Two caveats must accompany this comparison. First, COT is not scale-invariant: biological and physical scaling laws predict $COT \propto m^{-0.32}$ to $m^{-0.40}$ [31], [32], so larger platforms are expected to show lower mass-specific COT purely from Reynolds and volume-to-surface-area effects. Applying the allometric correction $(m/m_{ref})^{0.32}$ with $m_{ref} = 1.85$ kg raises BIOSwimmer's effective COT from 0.075 J/(N·m) to 0.21 J/(N·m). Even before correction, BIOSwimmer's reported value is already nearly double SwiftFin's 0.038 J/(N·m); the correction strengthens that gap and confirms that SwiftFin's advantage is not a scale artifact. Second, different groups use different tanks, sensors, and protocols; the same-body ablation of Section V provides the fairer transmission-to-transmission comparison. With these caveats, the cross-platform picture is consistent with

TABLE II

COMPARISON OF SWIFTFIN WITH EXISTING ROBOTIC FISH PLATFORMS. $U_{\max}$ AND $P_{\max}$ ARE AT THE MAXIMUM-SPEED CONDITION. **ALL COT_{min} VALUES REPRESENT EACH PLATFORM'S MINIMUM ACROSS ITS TESTED OPERATING ENVELOPE, DERIVED FROM THE AUTHORS' REPORTED DATA (SOURCE CONDITIONS IN THE FOOTNOTE), NOT NECESSARILY AT THE MAXIMUM-SPEED CONDITION.**

Platform	Year	L_b [m]	Mass [kg]	$U_{\max}$ [BL/s]	$P_{\max}$ [W]	COT _{min} [J/(N·m)]	η_{trans} [%]	Actuation
SwiftFin (this work) [†]	2026	0.520	1.85	4.40	2.16	0.038	96.0	Capstan
Chen <i>et al.</i> [7]	2022	0.277	0.380	3.80	36.1	5.66	~83	BLDC+gear+compliant
Wang <i>et al.</i> [10]	2025	0.160	—	0.28	—	—	—	DEA (soft)
Tunabot [1]	2019	0.255	0.306	4.00	2.5	0.817	~75	Crank
Tunabot Flex [2]	2021	0.255	0.310	4.60	2.1	0.590	~75	Crank
SPC-III [27]	2011	0.519	4.2	2.62	8.5	0.152	~82	Servo+linkage
OpenFish [28]	2022	0.522	1.1	1.63	3.2	0.349	~80	DC+gearbox
RoboTuna [20]	1999	1.24	23.0	1.00	45.0	0.161	~70	Cable-pulley
BIOSwimmer [29]	2012	1.52	45.0	1.50	75.0	0.075	~85	Servo
Hu <i>et al.</i> [30]	2015	0.70	5.5	2.86	15.0	0.139	~80	Servo

[†]SwiftFin: COT_{min} = 0.038 at $f = 2$ Hz, $U = 1.48$ m/s, $P = 1.02$ W; $U_{\max}$ at $f = 3$ Hz. Tunabot [1]: COT_{min} at $U = 1.0$ BL/s (Zhu *et al.*, Fig. 5). Tunabot Flex [2]: at $U \approx 1.2$ BL/s. Chen *et al.* [7]: at $U = 1.1$ BL/s (their Fig. 12). All other platforms: minimum COT extracted from published COT-versus-speed data.

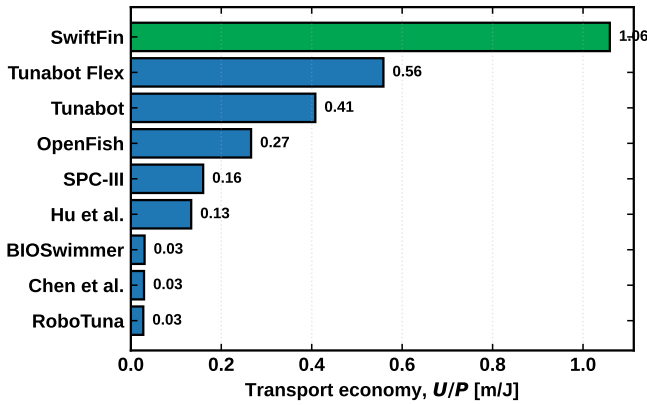


Fig. 23. Transport economy U/P (m/J) across comparison platforms. SwiftFin (green) leads the field at 1.06 m/J, a direct consequence of high transmission efficiency combined with moderate power draw.

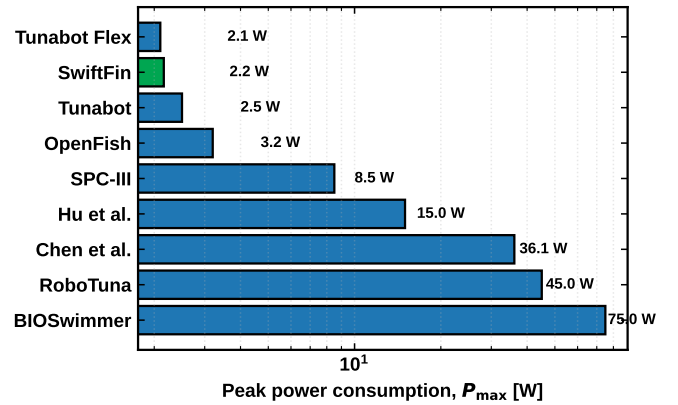


Fig. 24. Peak power consumption $P_{\max}$ (logarithmic scale) across comparison platforms. SwiftFin's 2.16 W is an order of magnitude below most comparable platforms.

the ablation: SwiftFin achieves the lowest COT in its scale class, and the identified mechanism (high transmission efficiency) is the same in both within-platform and across-platform views.

IX. CONCLUSION

A. Conclusion

This paper argues that transmission design should be treated as a primary hydromechanical design variable in robotic fish, not merely as an implementation detail. SwiftFin shows that when a low-backlash, high-efficiency transmission is paired with a body tuned for favorable fluid interaction, the resulting gains are not confined to bench-top drivetrain metrics; they propagate to system-level swimming economy.

Three design lessons emerge from the study. First, transmission efficiency matters at the same level as body shape and gait selection, because losses upstream of the tail directly raise the electrical cost of producing useful thrust. Second, backlash is not only a control nuisance: in oscillatory locomotion it creates a repeatable energetic penalty at every reversal, so reducing backlash improves both tracking fidelity and transport economy. Third, body design and actuation cannot be optimized

independently; the best performance arises when drivetrain, compliant tail, and hydrodynamic loading are co-designed as a coupled system.

From that perspective, the main contribution of SwiftFin is less the specific hardware embodiment than the design principle it validates: a backdrivable, nearly backlash-free transmission can materially shift the efficiency frontier of thunniform robots. We expect this principle to extend beyond the present platform to larger autonomous swimmers, multi-DOF fish bodies, and other oscillatory underwater robots in which repeated reversal and reflected inertia make transmission losses especially costly.

B. Limitations

Five limitations remain. First, the rigid anterior fuselage (60% of body length) limits maneuverability; turning radius and 3D motion were not characterized. Second, the capstan friction coefficient $\mu = 0.35$ was measured in the sealed dry compartment; water ingress would reduce μ to 0.05–0.07 and render the capstan inoperable, so hull integrity is essential. Third, the 200,000-cycle durability test represents only 28 h of continuous operation; endurance-class AUV missions require

testing to 10^6 cycles or more. Fourth, $k_{D,act}$ is fitted from COT data at $\theta_0 = 30^\circ$; direct wake flow-visualization would provide an independent check. Fifth, the servo-direct comparison conflates transmission, motor, and driver effects, so its 15% COT gap is not a pure transmission-only comparison.

C. Future Work

Four directions extend this work. **(i) Multi-DOF capstan drives.** The current capstan actuates a single yaw axis; the principle extends naturally to multiple DOF, with each axis retaining the efficiency and precision demonstrated here. Multi-DOF capstan-driven bodies with segment-level articulation would approach the three-dimensional agility biological fish exhibit. **(ii) Closed-loop kinematic optimization.** The capstan's independent amplitude and frequency adjustability, combined with low tracking error, enables central-pattern-generator (CPG) or model-predictive-control (MPC) controllers that adapt tail kinematics to minimize COT as speed or ambient flow changes. **(iii) Hydrodynamic validation.** Wake flow-visualization experiments would provide direct experimental verification of $k_{D,act}$ and η_h , testing the wake-recovery mechanism cited in this work. **(iv) Long-duration endurance.** Testing to 10^6 cycles or more, with particular attention to ultra-high-molecular-weight polyethylene (UHMWPE, the Dyneema fiber material) fiber fatigue under wet cyclic bending.

ACKNOWLEDGMENT

The authors thank the members of the laboratory for assistance with towing-tank experiments.

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